

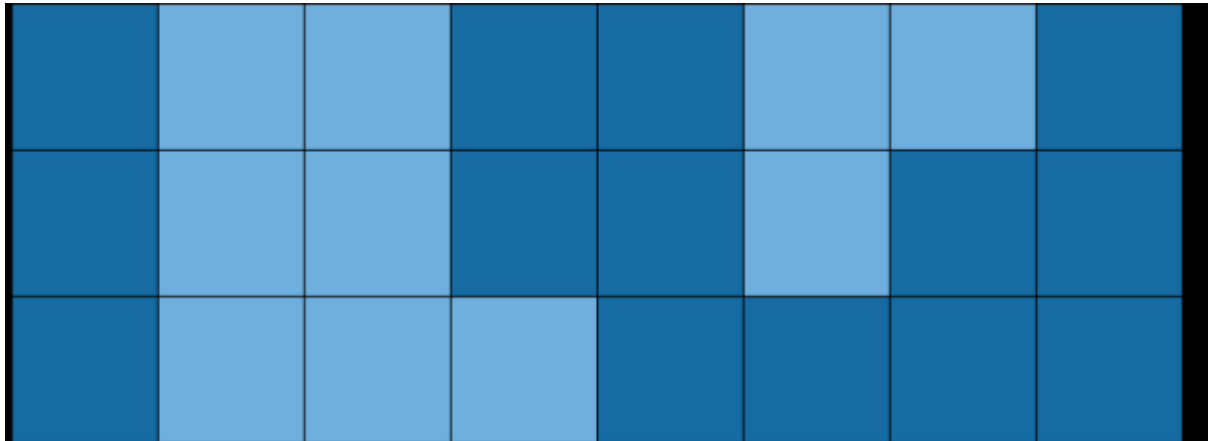
Title of the work: FDP Drone

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FDP 'drone'

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Introduction

The inefficient and delayed method of emergency response in urban areas has motivated the exploration of superior solutions.

Today, the FDP Drone (Firefighter, Doctor, Police Drone) project has evolved into a device that can intervene ahead of police, fire and medical crews to increase the chance of survival for individuals in danger. It can also help rescue crews by handing out first aid kits, communicating with victims by audio-visual communication, transmitting information to the pilot/dispatch in real time, and in the future, extinguishing low-intensity fires and chasing away aggressive animals using ultrasonic waves with systems under development.

Presentation of the objectives of the work

- Stimulate and encourage involvement in developing technical projects, solving complex tasks in a creative way;
- Stimulating and encouraging involvement in the development of projects with a positive social impact;
- Encouraging team spirit, sharing experiences and knowledge;
- Ability to increase victims' chances of survival by avoiding road traffic and can be controlled from anywhere in the world;
- Create a device with the ability to increase victims' chances of survival and inform authorities of what is happening so they can come prepared, sending live images and being able to communicate audio;
- Creating an educational STEAM-type framework (combining programming, design and mathematical skills).

Project summary

Objectives:

The FDP Drone (Firefighter, Doctor, Police) was created by applying computer science, mathematics, mechanics, electronics and programming skills. The aim of the drone is to intervene ahead of rescue crews to increase the chances of survival of certain people in danger. It can also assist rescue crews in high-intensity situations.

Motivation:

- The chance to accumulate knowledge, confidence and research, application and life skills;
- The chance to provide a more effective solution to combat accidents in urban areas;
- The chance to help people in danger.

Methods underlying the work

- STEAM is an educational concept based on the idea of educating students in five areas: science,

technology, engineering, design and mathematics, using a multidisciplinary and applied approach. STEAM integrates this knowledge into a coherent learning paradigm based on real-world applications, taking into account students' age and knowledge;

- Method of working with software and platforms by learning with demos and video tutorials;
- Discovery learning method.

Information about the work

The drone is a hexacopter based on a carbon fibre frame. It is equipped with six GARTT ML 5008 motors, which can rotate at up to 8800 RPM, and six 15-inch propellers. To control these motors, we use a Pixhawk 2.4.8 flight controller based on an STM32 microprocessor. Its main role is to take care of the control of the engines according to the data received from the user, with the help of the sensors implemented in it. To transmit data by radio about the movement of the drone we use a FlySky FS-i6X transmitter and a FlySky FS-iA10B receiver which is connected to the flight controller. On the drone we have mounted a flight computer (Raspberry Pi 4 Model B) based on a Broadcom BCM2711 processor. It communicates with the flight controller, in the case of long distance control, via the internet, and sends instructions to the flight controller, from the user, about the flight mode - manual (radio/internet control), autonomous (movement via GPS points), obstacle avoidance, person/object detection and tracking + facial recognition -, but it also sends information to the user about the state of the drone and the components mounted on it (GPS position, altitude, values from sensors, battery level etc.) . The system is equipped with 2 cameras, a radio camera (for transmitting live images over short distances) and a web camera (to observe images over the internet, over long distances and to use facial recognition, tracking and object/person detection).

This recognition uses the OpenCV library to detect faces and objects in a video stream and recognize them by using a pre-trained model. For the object recognition we used a library called YOLO, a popular object detection algorithm that can detect multiple objects in an image and identify their classes and bounding boxes, and for the facial recognition we are using another popular python library called 'facial_recognition', an open-source library that provides facial recognition capabilities. It is built on top of dlib, which is a popular C++ library for machine learning and computer vision. 'facial_recognition' uses a trained model with just a single picture (or more for even better results) to detect and recognize faces in images and video streams. The script captures frames from the camera and uses the face detection model to detect faces in the frames.

For audio communication with people in danger or people who can offer help we use a radio station. Four HC-SR04 ultrasonic sensors are mounted on the drone and are used to determine the distance of the drone from an object to avoid it, with the help of a library developed by us, which helps us to get a more accurate distance assessment with a temperature and humidity sensor. These are connected to an Arduino Nano board (based on the Atmega328p microcontroller) which sends the values to the Raspberry Pi board via serial communication. When the drone is in obstacle avoidance mode and the distance value measured by a particular sensor is less than or equal to a certain preset value (the minimum value at which an object can approach the drone), the Raspberry Pi sends instructions to the flight controller to control the motors so that the drone moves in the opposite direction of the approaching object to avoid it. The drone also comes equipped with MQ-4, MQ-5, MQ-6, MQ-8 and MQ-9 sensors that are used to detect certain gases. The gas detection sensors operate using an Arduino Nano board (based on the Atmega328p microcontroller) which sends the sensor values to the Raspberry Pi board, which then sends them to the user. The system comes equipped with a first-aid kit and eventually with a fire extinguishing ball.

Thus, by moving efficiently, transmitting real-time audio-visual information from the scene to the

pilot, extinguishing low-intensity fires, providing first-aid kits and avoiding obstacles, the FDP drone increases the chances of survival of people in danger.

Mode of operation

The FDP drone acts in critical situations early and arrives on the scene ahead of crews. There, it can connect live dispatch with victims, informing them what to do in their situation, detect certain types of gas to prevent possible explosions, hand out first aid kits, decrease the intensity of fire and bypass obstacles. In the future, we plan to add more functionality, such as detecting people trapped under rubble with a thermal camera and more (specified below).

Programming and software

We use the Mission Planner / QgroundControl application as software to observe and send data directly to the flight controller using the Raspberry Pi board. In short, the device on which the application is installed (laptop) is Port Forwarded over the internet, so that the Raspberry Pi can connect to it over the internet, acting as an intermediary between the flight controller and the device, making them communicate with each other through a communication protocol called MAVLink, so that the pilot/user can read data from modules and sensors connected directly to the flight controller (GPS, battery level sensor, atmospheric pressure sensor, gyroscope etc.), but also to set GPS waypoints for the drone to move autonomously and to control the drone manually via the internet, remotely, without a radio transmitter.

For the obstacle avoidance part, the microcontroller to which the four ultrasonic sensors are connected sends the values obtained from them to the Raspberry Pi via serial communication. Then, depending on the values received, the Raspberry Pi sends instructions to the flight controller via serial communication, so that it moves the motors in such a way as to avoid an object that gets too close to the drone, without the need for manual control.

For gas detection near the drone, the microcontroller on the Arduino Nano board reads the values from the gas sensors, processes them and sends them to the Raspberry Pi board via serial communication, which sends those values to the user via a server. The user reads those values through an interface that communicates with the server, the same interface that is used for the facial recognition, detection and tracking of an object/person, all done over the internet. We chose to process information by streaming over to a server has several advantages. **First**, a powerful PC typically has more processing power and memory than a Raspberry Pi, allowing it to handle more complex tasks and process larger amounts of data. This is particularly useful for tasks that require real-time processing, such as image or video recognition. Second, streaming the data to a powerful PC allows for greater flexibility in terms of the software and algorithms that can be used. Many advanced machine learning and AI algorithms require significant processing power and memory to run, which may not be available on a Raspberry Pi. By streaming the data to a powerful PC, it is possible to take advantage of more powerful algorithms and software libraries that can help to improve the accuracy and performance of the system.

For object recognition we are using Yolo, a popular object detection and image segmentation model developed by Joseph Redmon and Ali Farhadi at the University of Washington. YOLOv5 was trained using a different approach than previous versions of YOLO. YOLOv5 used a single-stage detection model with a novel approach called anchor-free detection. In the traditional approach to object detection, bounding boxes are predicted using predefined anchors or templates. These anchors are used to predict the size and location of objects in an image. However, YOLOv5 used an anchor-free approach, which does not require predefined anchors. Instead, YOLOv5 used a regression-based

approach to predict the location and size of objects in an image. This approach is based on the center point of an object, and predicts the offset of the object's center point from the grid cell that contains it. The model also predicts the width and height of the object relative to the grid cell size. To train the YOLOv5 model, a large dataset of annotated images was used. This dataset was used to train the model using a process called backpropagation. Backpropagation involves iteratively adjusting the weights of the model to minimize the difference between the predicted output and the ground truth labels. YOLOv5 also used several training techniques to improve the accuracy and speed of the model. These techniques included label smoothing, data augmentation, and mixed precision training. Label smoothing involves adding a small amount of noise to the ground truth labels, which helps to prevent the model from becoming overconfident in its predictions. Data augmentation involves applying random transformations to the training images, such as flipping or rotating, to increase the variability of the training data. Mixed precision training involves using a combination of lower-precision and higher-precision calculations to reduce the memory requirements and improve the speed of training. Overall, YOLOv5 was trained using a combination of novel techniques and advanced training methods to improve the accuracy and speed of the model for object detection tasks. At the same time, the 'facial_recognition' library uses a face detection algorithm to locate the faces in an input image or video frame. This algorithm uses a combination of Haar cascades and deep learning to detect faces, and can also detect multiple faces in a single image or frame. Once the faces have been detected, the library uses a deep learning algorithm to extract facial features from each face. These facial features include things like the distance between the eyes, the shape of the nose, and the curvature of the lips. The facial features are then compared to a database of known faces using a distance metric. This distance metric measures the similarity between the features of the input face and the features of each known face in the database. The library then returns the identity of the closest match. The database of known faces can be created by manually adding images of known individuals or by using an automated process to extract faces from a large dataset of images. The library also includes functionality for creating and managing these databases. It's

Experimental results

- Ability to communicate with victims, informing them about what they should do in their situation;
- Assisting crews and helping victims;

Conclusion

The FDP Drone system developed to intervene in certain emergency situations can be improved by introducing and implementing new functionalities such as:

- Detection of people trapped under rubble using a thermal camera;
- The ability to chase away aggressive animals with an ultrasound gun;
- Implementation of the anti-fire ball throwing mechanism;
- Drone charging and housing station.

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- <https://www.raspberrypi.org/>
- <https://dronekit.io/>
- <http://mavlink.io/en/>
- <https://opencv.org/>
- <https://github.com/SpulberGeorge/EasyUltrasonic>
- <https://github.com/ultralytics/yolov5>
- <https://pypi.org/project/face-recognition/>

Annexes

Annex 1: Components list

- Carbon fibre frame
- GART ML 5008 BLDC Motors [x6]
- ESC (Electronic Speed Controller) 40A [x6]
- 15-inch propellers [x6]
- PixHawk 2.4.8 flight controller
- M8N GPS module
- FlySky FS-iA10B radio receiver
- FlySky FS-i6X radio transmitter
- Raspberry Pi 4 Model B
- WEB Camera
- Radio camera + transmitter
- Camera Gimbal
- MQ-4 sensor (for methane gas detection)
- MQ-5 sensor (for detecting coal smoke)
- MQ-6 sensor (for butane gas detection)
- MQ-8 sensor (for hydrogen detection)
- MQ-9 sensor (for carbon monoxide detection)
- Ultrasonic sensors HC-SR04 (for determining the distance to an object) [x6]
- Arduino Nano board for gas sensors
- Arduino Nano board for ultrasonic sensors
- First aid kit
- ElideFire anti-fire ball

Annex 2: Help

In carrying out our project, we enjoyed the experience and guidance of Ms. Întuneric Ana who helped us to put into practice the ideas and knowledge gained.

Annex 3: Schematics / Pictures

The old versions of the drone



The new versions of the drone



Annex 4: Contact

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